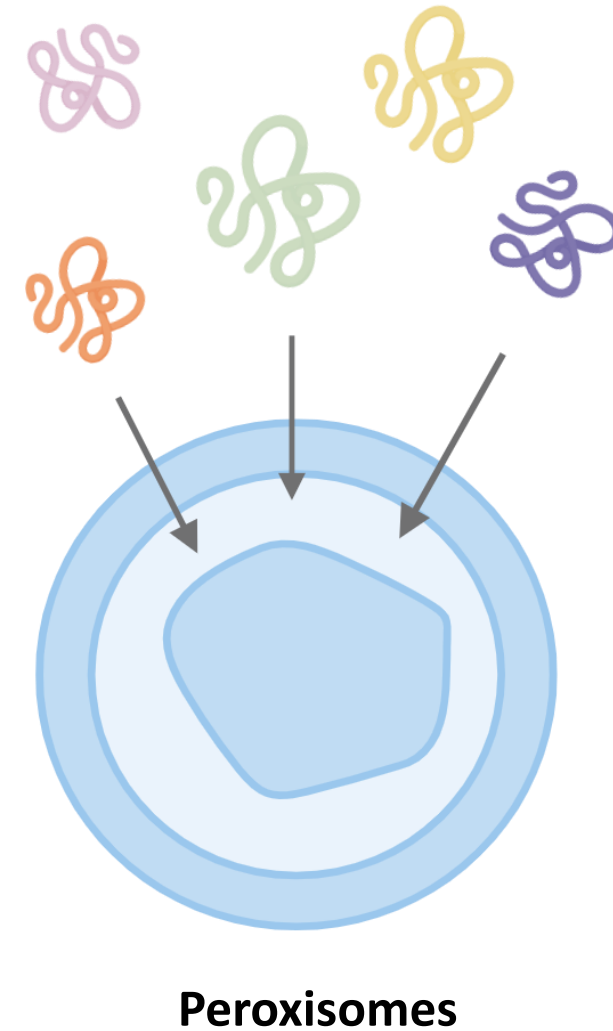
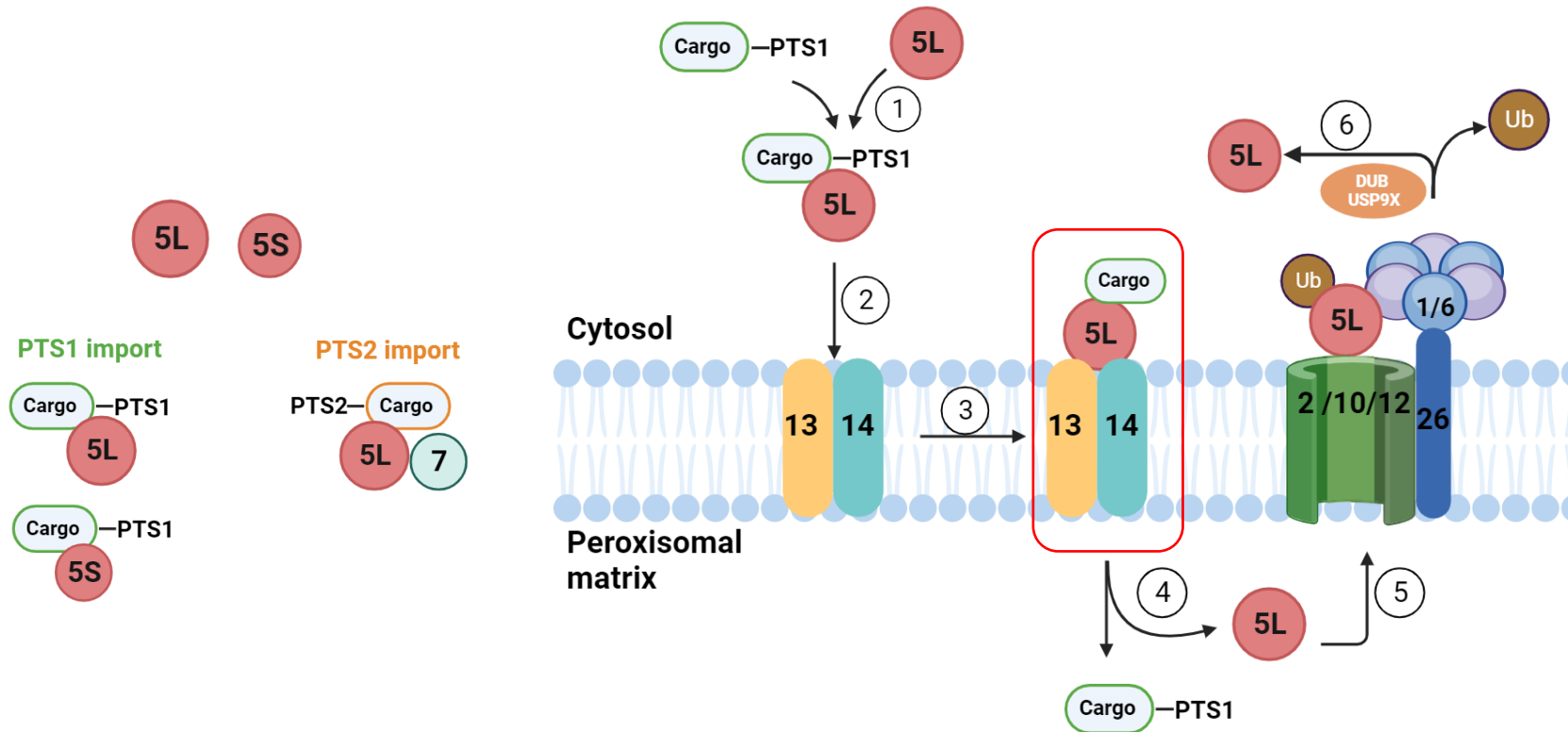


Dynamics of the translocation pore of the human peroxisomal protein import machinery



Peroxisomal protein import in humans



Question:

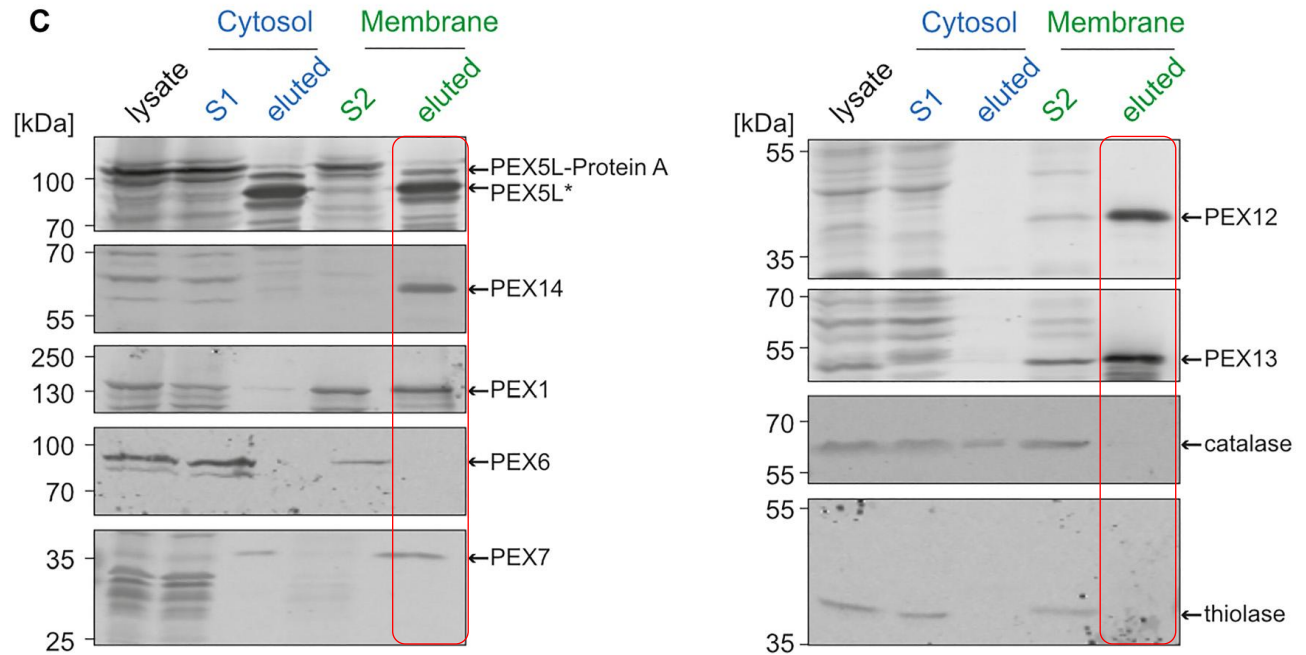
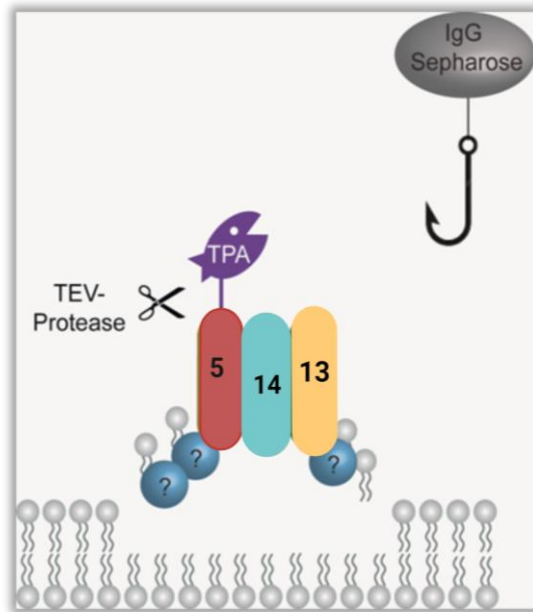
Does the human import machinery, particularly the interaction between PEX5L and PEX14, exhibit pore-forming activity?

How does the incubation with cargo regulate channel activity?

Aim and approach:

Reconstitution of PEX5L complexes into LUVs and perform electrophysiological characterization

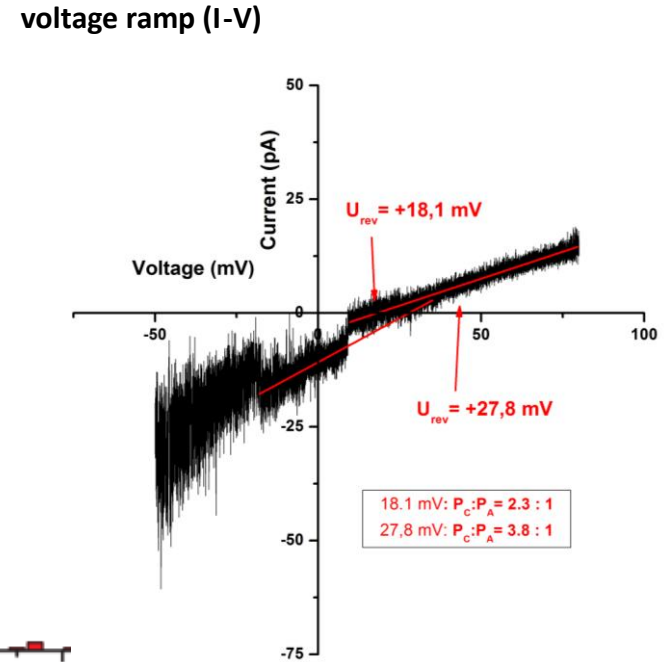
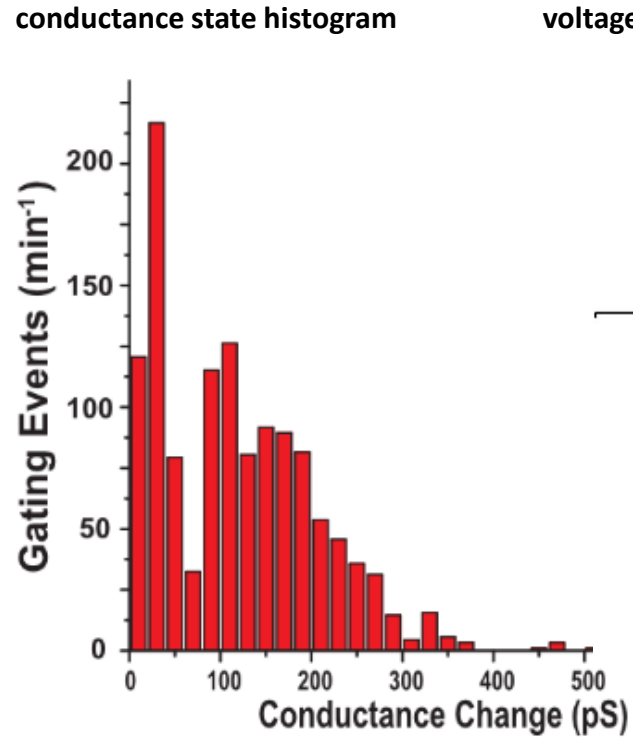
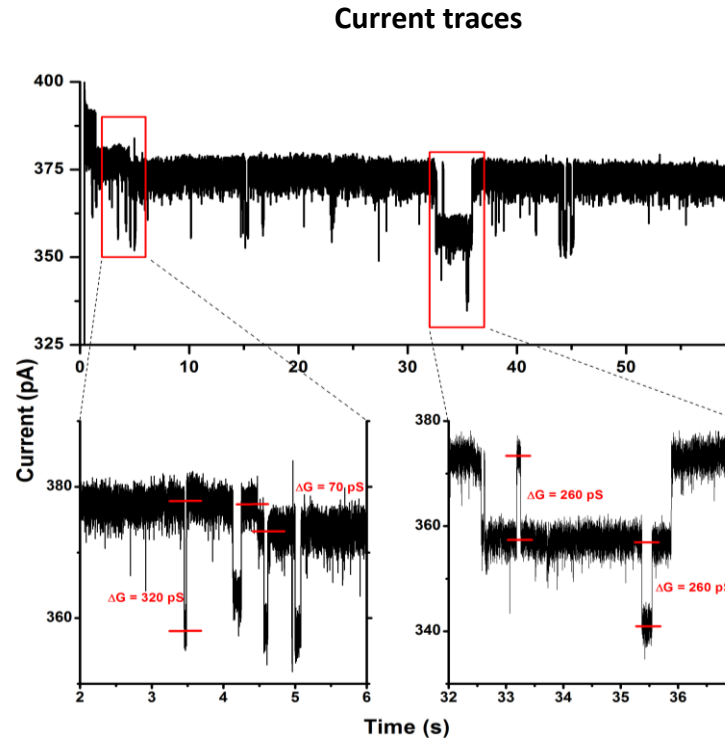
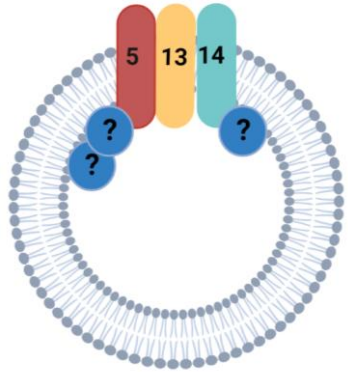
Isolation of Human PEX5L-complexes



Isolation and Analysis by M. Reuter and J. Klümper, Bochum

Ghosh et al., 2022

Electrophysiology of human PEX5L complexes



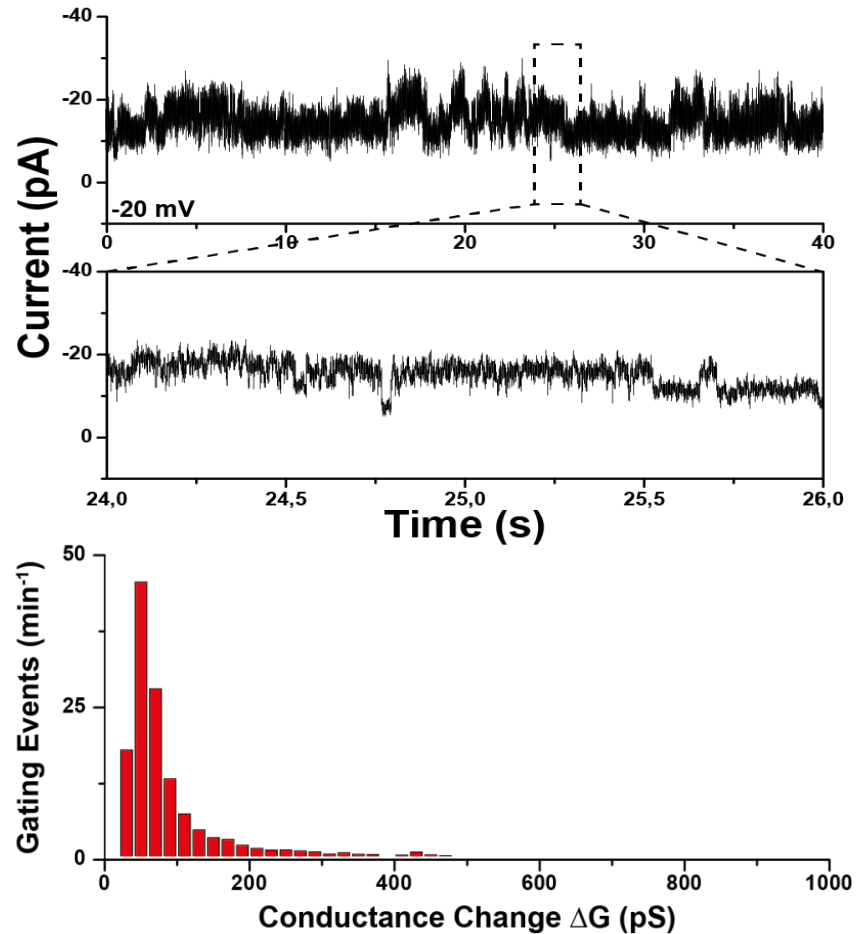
Ghosh et al., 2022

- Dynamic gating
- Pattern of conductance states
- Mild but changing ion selectivity

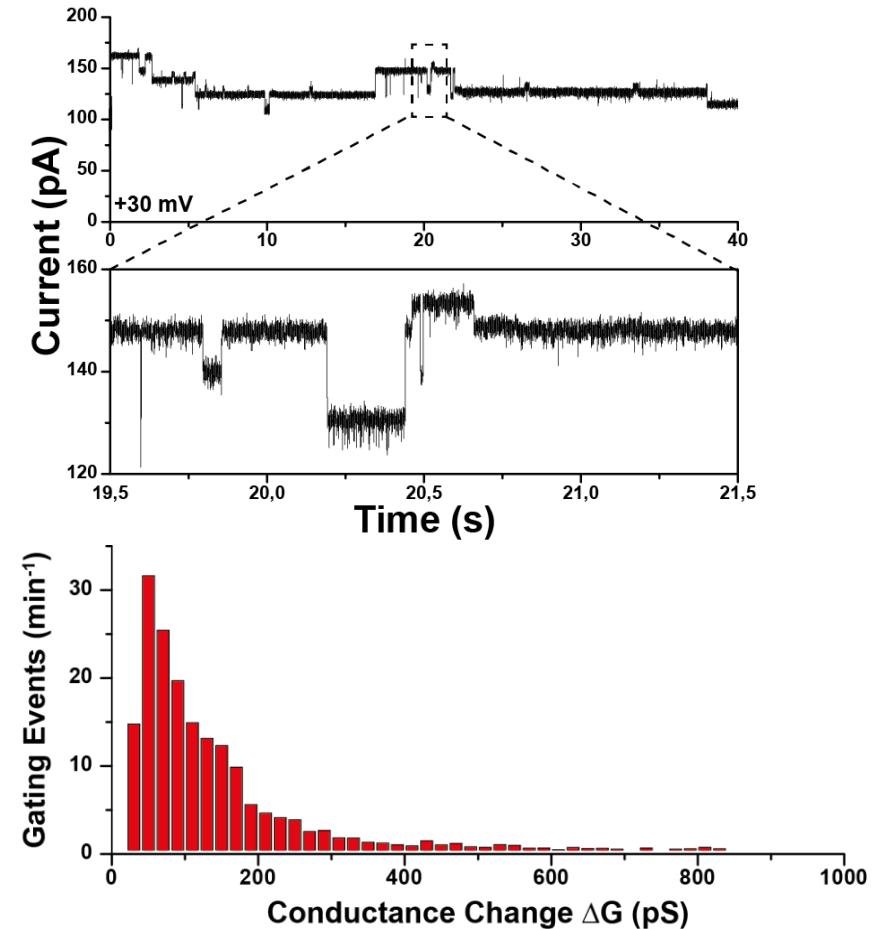
**Human PEX5L-complexes display
similar channel properties as yeast PTS1 import pore**

Low and high -conductance pore in PEX5L-complexes

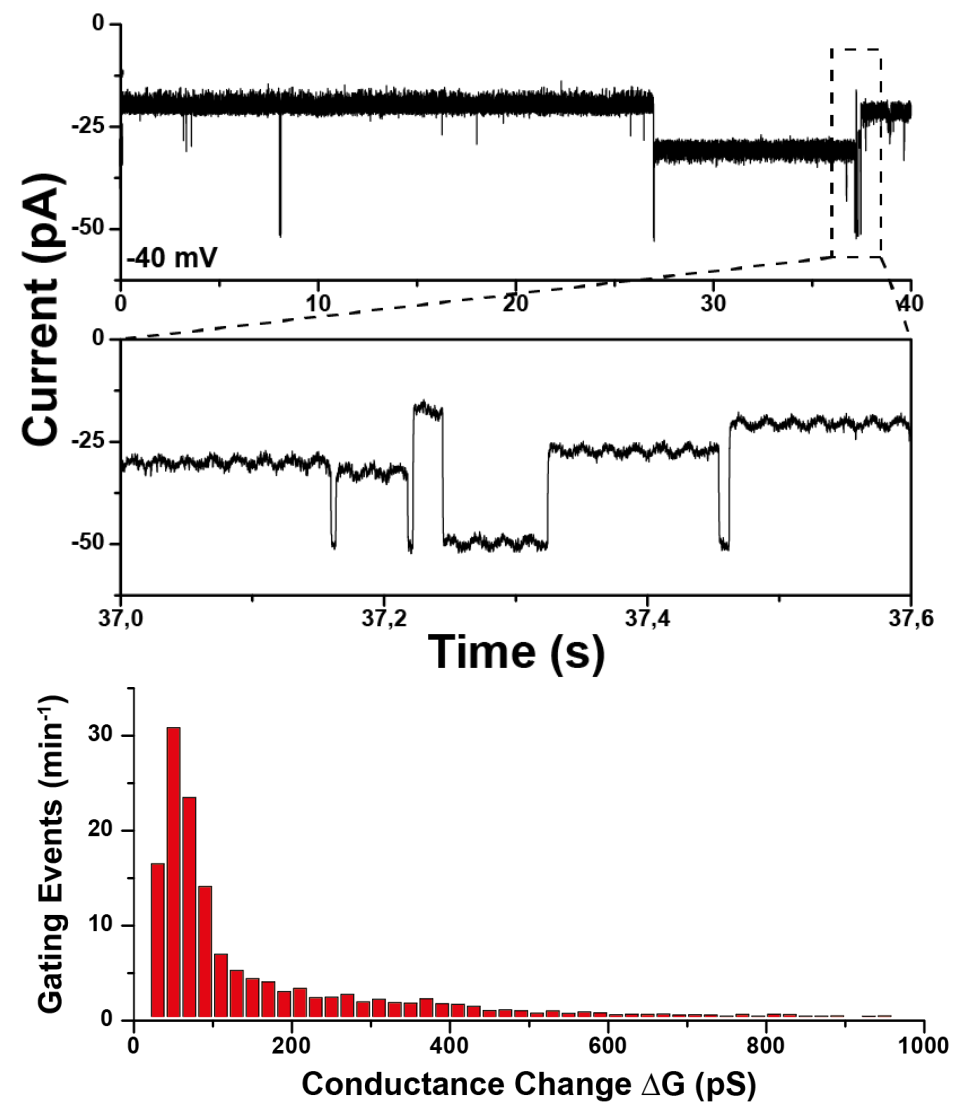
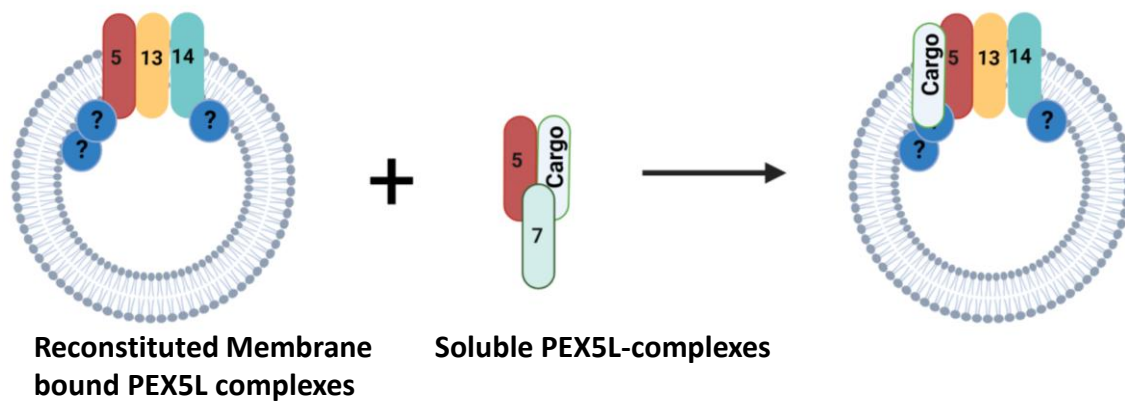
Low-conductance pore



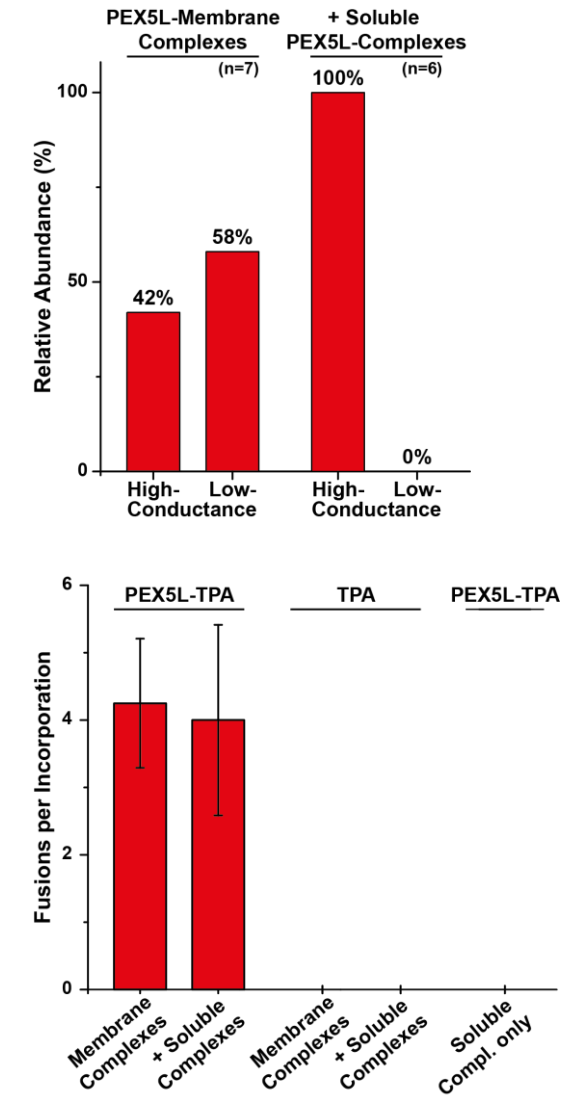
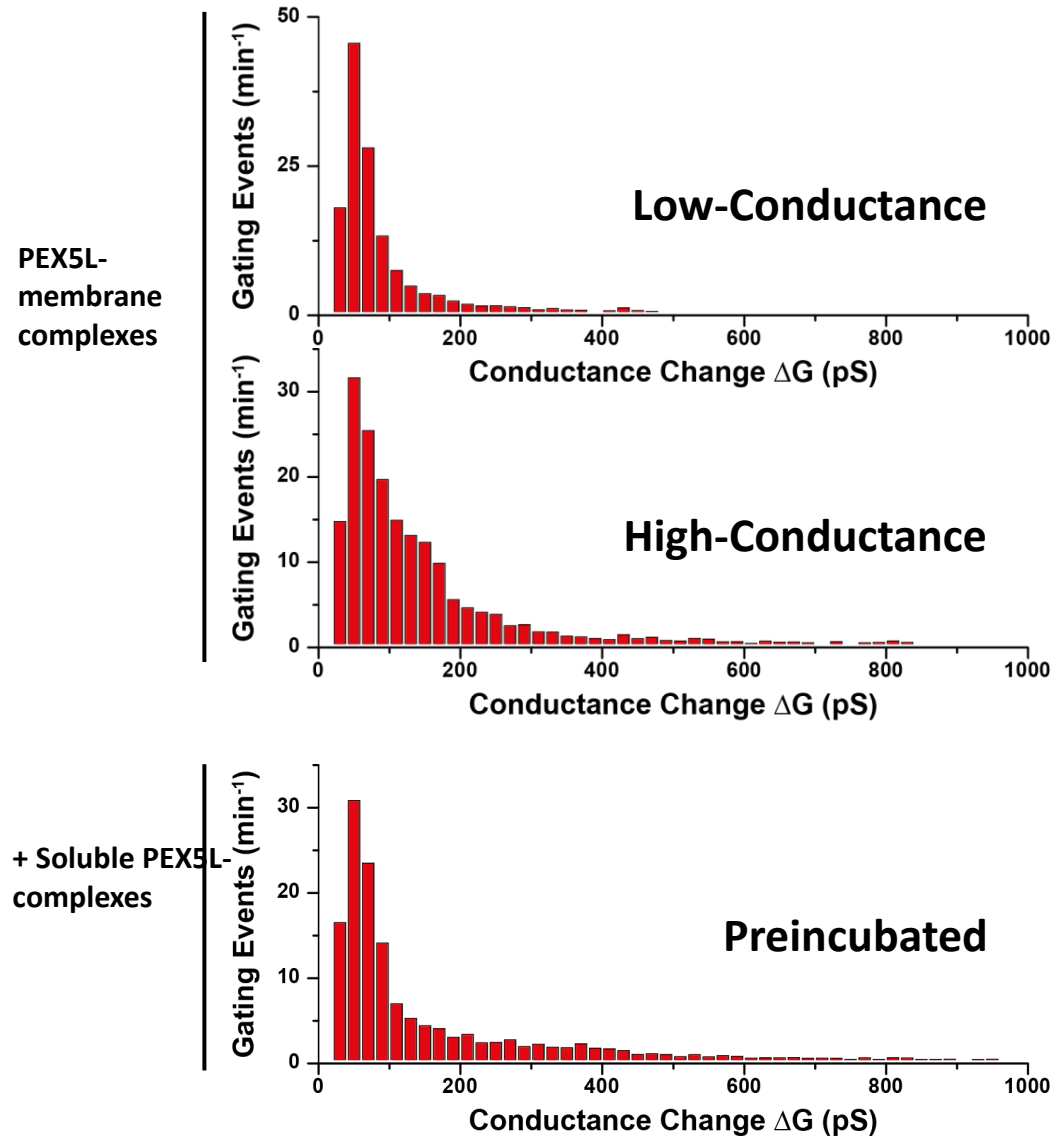
High -conductance pore



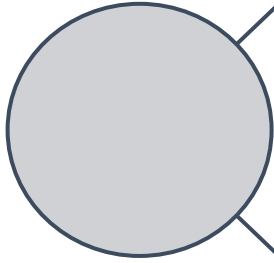
Preincubation with soluble PEX5L-complexes



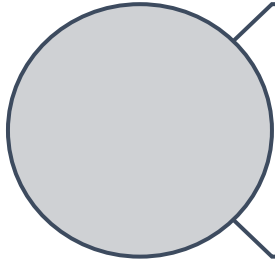
Preincubation with soluble PEX5L-complexes



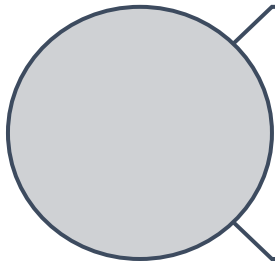
Summary



Human PEX5L has pore forming capabilities resembles to that of yeast Pex5 pore



Two PEX5L-populations: low- and high-conductance with voltage dependent gating and mild ion selectivity



Incubation with receptor-cargo complexes stimulates transition from low-conductance to high-conductance, opening the pore